Message-Level Communication Functions in LLM Multi-Agent Systems: A Taxonomy and Empirical Study

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Abstract

Removing the least informative messages from LLM multi-agent conversations produces the largest behavioral disruption. Across 2,693 GSM8K traces spanning three backbones (Qwen2.5-7B, Llama-3.1-8B, Gemma-2-9B) and three communication topologies, counterfactual ablation reveals that excising simple acknowledgments—informationally sparse messages such as "I agree" or "Let me check"—changes downstream behavior far more (mean similarity = 0.35) than removing content-rich reasoning guidance (0.78). Individual-message ablation confirms this: a single removed acknowledgment alters the final answer in 66% of cases versus 29% for reasoning guidance. We trace this counter-intuitive pattern to three qualitatively distinct, backbone-dependent communication modes: *function-driven* (Qwen; message type determines impact, $F = 232$, $p < 0.001$), *position-driven* (Gemma; impact depends on conversational position, not content, $R^2 = 0.34$), and *undifferentiated* (Llama; $R^2 = 0.02$). A crossover interaction between backbone and topology shows that dynamic routing disrupts endogenous structure in function-driven models while activating it in undifferentiated ones. The impact gradient is robust across four similarity metrics (Spearman $\rho > 0.91$), temperature settings, and a MATH cross-domain validation ($p = 0.011$). These results provide empirical evidence against content-based message pruning heuristics and establish that communication structure in multi-agent systems is an emergent, model-dependent property rather than a universal feature of collaborative interaction.

1. Introduction

Communication optimization in LLM-based multi-agent systems rests on an implicit assumption: message value correlates with information content. Pruning methods [Xing et al.(2025), Zhang et al.(2025c), Zhang et al.(2025b)] discard messages deemed redundant or low-content; routing methods [Li et al.(2026), Chen et al.(2026a)] direct information-rich messages to relevant agents. This assumption is intuitive—surely a detailed reasoning trace matters more than a brief "I agree"—but it has never been empirically tested at the message level.

We test it, and find the opposite. Through counterfactual ablation of 2,693 multi-agent GSM8K traces across three backbones and three topologies, we show that removing informationally sparse acknowledgments produces the *largest* behavioral disruption (mean similarity = 0.35), while removing content-rich reasoning guidance produces the *smallest* (0.78). A single excised acknowledgment alters the final answer in 66% of cases, versus 29% for reasoning guidance. Content-based pruning heuristics would discard the wrong messages.

Why do low-content messages carry high impact? We identify three qualitatively distinct communication modes that explain this pattern:

1. **Impact gradient and structural load-bearing** (C1). Under fixed-turn topologies, the per-type impact gradient is inversely related to information content. Acknowledgments are informationally sparse but contextually load-bearing: their absence violates conversational expectations, forcing large behavioral shifts. The neutralize ablation—which preserves message structure while removing content—produces the opposite gradient, confirming that acknowledgments' high impact stems from structural presence rather than informational content (Appendix E). This gradient is robust across four independent similarity metrics (Spearman $\rho > 0.91$; Appendix G).
2. **Three backbone-dependent communication modes** (C2). OLS ANCOVA reveals that the impact gradient is not universal but backbone-dependent. Qwen exhibits *function-driven* communication ($F = 232$, $p < 0.001$): message type counterfactually determines downstream impact. Gemma exhibits *position-driven* communication ($R^2 = 0.34$, function type $F = 0.43$, n.s.): impact varies with conversational position, not communicative content. Llama exhibits *undifferentiated* communication ($R^2 = 0.02$): messages have interchangeable effects. These are qualitatively distinct modes, not points on a gradient.
3. **Crossover interaction** (C3). Selector topology partially converges the function-driven and undifferentiated modes: Qwen's R^2 drops from 0.45 to 0.065 while Llama's rises from 0.02 to 0.056 (backbone $\times$ topology $F = 181$, $\eta^2 = 0.043$). Dynamic routing provides a structural scaffold that is redundant for capable models but compensatory for weaker ones.

Cross-domain validation on MATH confirms the function-driven mode (Qwen $F = 3.77$, $p = 0.011$) and position-driven mode (Gemma); temperature robustness checks ($T = 0.7$) confirm stable results (Appendices H, C).

Our approach combines a five-category functional taxonomy (INFORMATION PROVISION, REASONING GUIDANCE, VERIFICATION, COORDINATION, ACKNOWLEDGMENT) validated via cross-model annotation ($\kappa = 0.35$–0.67) with counterfactual ablation that isolates each message's downstream behavioral impact. Theoretical work [Rizvi-Martel et al.(2026)] establishes *when* communication helps; trace-level diagnosis [Cemri et al.(2025)] identifies *what went wrong*. We provide the missing *what*: a functional characterization of message-level contributions that reveals communication structure as an emergent, model-dependent property—not a universal feature of multi-agent interaction.

2. Related Work

Our work draws on four threads: theoretical foundations, LLM-based MAS, communication efficiency, and message function analysis.

Theoretical foundations. Rizvi-Martel *et al.* [Rizvi-Martel et al.(2026)] derive information-theoretic bounds on when multi-agent communication helps, establishing *when* communication is necessary but not *what* content the bandwidth should carry. The Five Ws survey [Chen et al.(2026b)] identifies "what to communicate" as the most understudied dimension. Our taxonomy directly fills this gap.

LLM-based MAS. Frameworks such as CAMEL [Li et al.(2023)Li, Hammoud, Itani, Khizbullin, and Ghanem], MetaGPT [Hong et al.(2024)Hong, Zhuge, Chen, Zheng, Cheng, Zhang, Wang, Wang, Yau, Lin, et al.], AG2 [Wu et al.(2024)Wu, Bansal, Zhang, Wu, Li, Zhu, Jiang, Zhang, Zhang, Liu, Awadallah, White, Burger, and W and AgentVerse [Chen et al.(2024)Chen, Su, Zuo, Yang, Yuan, Chan, Yu, Lu, Hung, Qian, et al.] demonstrate that inter-agent communication affects performance. Multi-agent debate [Du et al.(2024)Du, Li, Torralba, improves reasoning, though majority voting may largely account for the gains [Smit et al.(2025)]; controlled studies confirm that intrinsic reasoning strength dominates structural parameters [Wu et al.(2025)Wu, Li, and Li], and deliberation protocols can underperform simple best-of-N selection [de Font-Reaulx et al.(2026)]. Yan *et al.* [Yan et al.(2025)Yan, Zhou, Zhang, Zhang, Zhou, Miao, provide a communication-centric survey organizing MAS research by architecture, strategy, and content, identifying message-level empirical analysis as an open direction. These studies

Table 1: Taxonomy of communication function types with definitions and examples from GSM8K solver–checker dialogues.

Function Type	Definition	Example
Information Provision	Supplies factual content, data, or intermediate results	"The total cost is $3 \times \$5 = \$15$ for the shirts."
Reasoning Guidance	Directs reasoning through decomposition or strategic suggestions	"First calculate the weekly cost, then multiply by 4."
Verification	Checks or corrects the correctness of prior messages	"$17 \times 3 = 51$, not 53. The corrected answer is \$51."
Coordination	Manages dialogue flow, assigns roles, or delegates tasks	"I'll solve first, then you verify step by step."
Acknowledgment	Simple confirmations or agreements with no new content	"Yes, that's correct. The answer is 42."

establish that communication matters without characterizing the mechanisms through which it operates.

Communication efficiency and topology. AgentDropout [Xing et al.(2025)], AgentPrune [Zhang et al.(2025b)], AGP [Zhang et al.(2025c)], SupervisorAgent [Lin et al.(2026)], and OPTIMA [Chen et al.(2025)] achieve 20–90% token reduction by pruning messages, but treat message utility as a scalar without characterizing qualitative differences. ARG-Designer [Li et al.(2026)] and GoAgent [Chen et al.(2026a)] optimize topologies; information propagation studies [Jia et al.(2025)] use a binary correct/erroneous taxonomy. Our work provides the functional categories that efficiency and topology methods currently lack, and studies the backbone–topology interaction invisible to topology-only analyses.

Communication strategy analysis. Most closely related to our work, Zhang *et al.* [Zhang et al.(2025a)Zhang, Zhai analyze dialogue acts in collaborative LLM agents on mathematical problem-solving, finding that peer-to-peer collaboration achieves the highest accuracy. Our work differs in three respects: (1) we measure *counterfactual impact* through remove/neutralize ablation rather than descriptive dialogue act frequencies; (2) we provide statistical quantification via ANCOVA decomposition (F-statistics, R^2); and (3) we discover a backbone $\times$ topology crossover interaction invisible to single-backbone analyses.

Attribution and message function. AgentSHAP [Horovicz et al.(2025)] quantifies tool importance via Shapley values at the agent level. MAST [Cemri et al.(2025)] identifies 14 failure modes from 1,600+ traces but operates at the execution-trace level, not the message level. Our ablation framework provides finer-grained (individual message) and qualitatively richer (function type) analysis. Classical speech act theory [Austin(1962), Searle(1969)], FIPA ACL [Foundation for Intelligent Physical Agents(2002)], and dialogue act taxonomies [Core and Allen(1997), Bunt et al.(2012)Bunt, Alexandersson, Choe, Fang, Hasida, Petukhova, Popes inform our categories, but our taxonomy is empirically grounded in LLM agent behavior and defined by measurable counterfactual impact—a dimension absent from classical frameworks.

3. Taxonomy of Communication Functions

We define five exhaustive, mutually exclusive communication function types (Table 1). Ambiguity is resolved by hierarchical priority: *coordination* $\succ$ *verification* $\succ$ *reasoning_guidance* $\succ$ *information_provision* $\succ$ *acknowledgment*, reflecting the principle that structurally higher-level functions take precedence.

Information provision supplies facts that expand the shared state without directing how they should be used. Reasoning guidance is prescriptive: it tells the recipient *how* to

proceed. VERIFICATION assesses prior contributions' correctness. COORDINATION manages meta-structure (roles, turn-taking) without contributing task content. ACKNOWLEDGMENT confirms receipt or agreement; while informationally sparse, it maintains conversational coherence.

Annotation protocol. Each message is annotated by an LLM annotator using a structured prompt containing taxonomy definitions, decision rules, and three in-context examples per type, with temperature 0 and constrained output. Cross-model validation uses two independent annotator models per message; we adopt $\kappa \geq 0.4$ as our target threshold for cross-model agreement and report Cohen's κ throughout (Section 6.4).

4. Methodology

4.1. Multi-Agent System Setup

We use AG2 (AutoGen) [Wu et al.(2024)Wu, Bansal, Zhang, Wu, Li, Zhu, Jiang, Zhang, Zhang, Liu, Awadallah, Wh with solver–checker agent teams across three backbones of comparable scale: Qwen2.5-7B-Instruct [Yang et al.(2024)Yang, Yang, Zhang, Hui, Zheng, Yu, Li, Liu, Huang, Wei, et al.], Llama-3.1-8B-Instruct [Grattafiori et al.(2024)Grattafiori, Dubey, Jauhri, Pandey, Kadian, Al-Dahle, Letman, Mathu and Gemma-2-9B-IT [Gemma Team(2024)] (7–9B parameters). We study three topologies: **round-robin** (fixed cyclic order), **sequential** (strict two-phase rounds), and **selector** (dynamic routing via a selector agent). All models are served via vLLM [Kwon et al.(2023)Kwon, Li, Zhuang, Sheng, Zhen with greedy decoding (temperature = 0, max tokens = 2048). For each of the $3 \times 3 = 9$ configurations, we collect traces on 300 GSM8K [Cobbe et al.(2021)Cobbe, Kosaraju, Bavarian, Chen, Jun, Kaiser, Plappert, test-split tasks sampled with seed = 42, yielding 2,693 non-empty traces. The MATH cross-domain validation (Appendix H) uses 50 tasks at Level 3–5 (Algebra and Number Theory), also sampled with seed = 42. The full annotation prompt is reproduced in Appendix I.

4.2. Message-Level Ablation

To isolate each message's counterfactual effect on downstream behavior, we introduce two ablation types. **Remove ablation** deletes message m_i from trace $\mathcal{T}$ and regenerates m_{i+1}. **Neutralize ablation** replaces m_i with "`[message removed]`", preserving positional structure while removing content. Comparing the two disentangles content effects from structural presence effects [McNemar(1947)].

We measure impact as:

$$\text{sim}(m_i) = \cos\big(\mathbf{e}(m_{i+1}),\, \mathbf{e}(m'_{i+1})\big) \tag{1}$$

where $\mathbf{e}(\cdot)$ is a sentence embedding (`all-MiniLM-L6-v2`; [Reimers and Gurevych(2019)]). Lower similarity indicates higher message impact. We report similarity throughout; impact $= 1 - \text{sim}$.

4.3. Statistical Framework

We test whether function types produce different impact distributions using the Kruskal–Wallis H test [Kruskal and Wallis(1952)] (non-parametric, appropriate for bounded, skewed distributions), with Dunn's post-hoc tests [Dunn(1964)] and Bonferroni correction.

To control for the position confound—function types are non-uniformly distributed across conversation positions—we fit an OLS ANCOVA:

$$\text{sim}(m_i) = \beta_0 + \sum_{j=1}^{4} \beta_j \cdot \mathbb{1}[\text{type}(m_i) = j] + \gamma_1 \cdot \text{pos}(m_i) + \gamma_2 \cdot \text{round}(m_i) + \epsilon_i \tag{2}$$

The partial F-statistic for the function type factor tests whether function type effects persist after controlling for position and round number. We report HC3 heteroscedasticity-consistent standard errors throughout; residual diagnostics are in Appendix D. We further validate with a pooled three-way ANOVA across all backbones and topologies (Section 6.4).

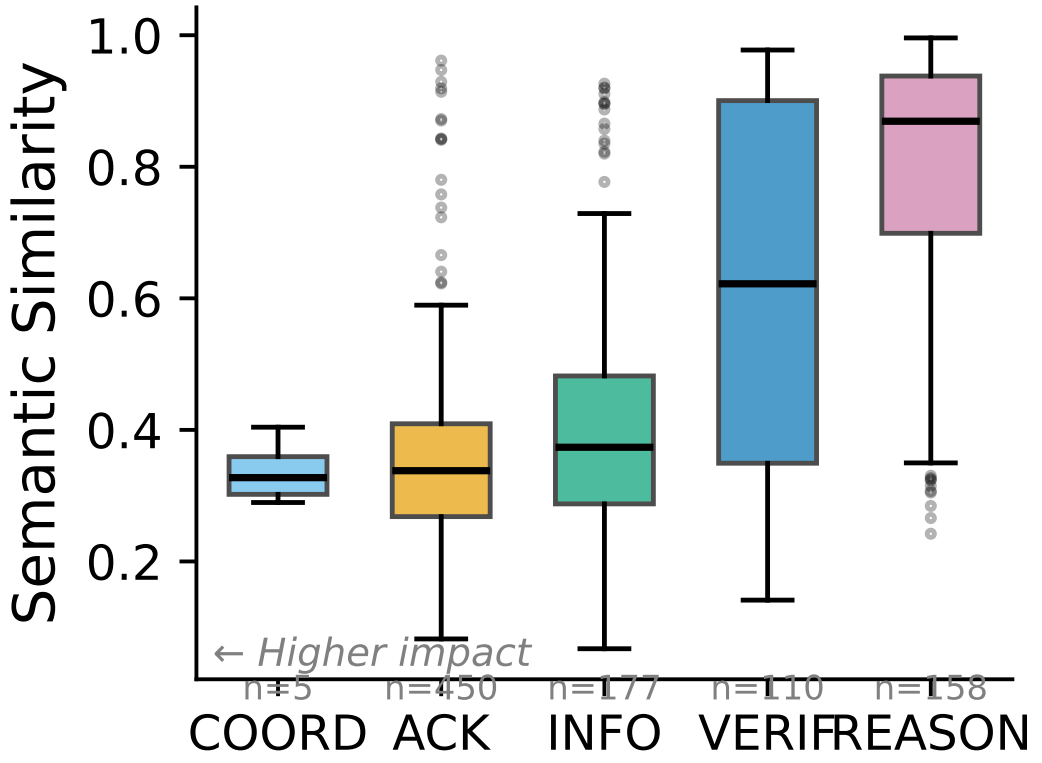


Figure 1: Semantic similarity by function type (Qwen, round-robin). Lower similarity = higher impact. Acknowledgment messages are the most impactful despite carrying the least information.

5. Experiments

Each agent team uses AG2's `ConversableAgent` with task-specific system prompts (solver: step-by-step solutions; checker: verify and flag errors). Under selector topology, a third agent routes messages via `GroupChat` with `speaker_selection_method="auto"`. Conversations are limited to 10 rounds. Function type labels are assigned by two independent LLM annotators (Qwen2.5-7B as Model A, Llama-3.1-8B as Model B) with structured JSON output. Solution correctness is determined by comparing extracted numerical answers against GSM8K ground truth.

6. Results

We report results across all nine backbone–topology configurations.

6.1. Counter-Intuitive Impact Gradient

For Qwen under fixed-turn topologies (Figure 1), removing ACKNOWLEDGMENT messages—the least informative category—produces the largest behavioral change (mean similarity $= 0.352$, $n = 450$), followed by INFORMATION_PROVISION (0.410, $n = 177$), VERIFICATION (0.616, $n = 110$), and REASONING_GUIDANCE (0.778, $n = 158$). The impact gradient is inversely related to information content: informationally sparse acknowledgments are more impactful than content-rich reasoning guidance.

Pruning evidence. Individual-message ablation confirms this gradient at the single-message level: removing one acknowledgment alters the final answer in 66% of cases, versus 29% for reasoning guidance. The neutralize ablation—which preserves message structure

Table 2: Kruskal–Wallis H statistics for semantic similarity across function types. All $p < 0.01$. Group counts vary (4–5 under RR/SEQ, 2–3 under selector); H values not directly comparable across topologies.

Backbone	**RR**	**SEQ**	**Selector**
Qwen2.5-7B	287.4	288.1	50.8
Gemma-2-9B	21.5	25.1	6.9
Llama-3.1-8B	13.0	15.8	63.8

Table 3: OLS ANCOVA: partial F for function type (controlling for position and round) and full-model R^2. $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; n.s. = not significant.

	RR		**SEQ**		**Selector**	
Backbone	F	R^2	F	R^2	F	R^2
Qwen2.5-7B	232^{***}	.450	225^{***}	.446	55.8^{***}	.065
Gemma-2-9B	$0.43^{\text{n.s.}}$	.341	$0.44^{\text{n.s.}}$	.333	$0.60^{\text{n.s.}}$	.036
Llama-3.1-8B	3.81^{*}	.021	4.45^{**}	.020	17.3^{***}	.056

while removing content—produces the opposite gradient: information-rich types suffer most. This double dissociation confirms that acknowledgment's high remove-impact stems from structural presence rather than informational content (Appendix E). The gradient is robust across four independent text similarity metrics (Spearman $\rho > 0.91$; Appendix G) and across RR and SEQ topologies, but breaks down under selector topology.

6.2. Three Communication Modes

The Kruskal–Wallis H test confirms that function type significantly predicts message impact across all configurations ($p < 0.01$), but the magnitude varies by an order of magnitude across backbones (Table 2). However, this omnibus test conflates function type effects with positional confounds.

Position control via OLS ANCOVA. To disentangle function type from positional confounds, we fit the ANCOVA model (Equation 2). Table 3 reveals three qualitatively distinct communication modes that explain *why* the impact gradient exists for some backbones but not others.

- **Function-driven** (Qwen): function type is the dominant predictor ($F = 232$), driving most of the $R^2 = 0.45$. The impact gradient (§6.1) is a direct consequence: because function type counterfactually determines impact, removing an entire functional category produces category-specific behavioral changes.
- **Position-driven** (Gemma): $R^2 = 0.34$ but function type $F = 0.43$ (n.s.)—all explained variance comes from message position and round number. Adding a quadratic position term absorbs $\Delta R^2 = 0.228$, rendering function type non-significant ($p = 0.806$; Appendix D), confirming that Gemma's impact variation is driven entirely by nonlinear positional dynamics.
- **Undifferentiated** (Llama): $R^2 = 0.02$; neither function type nor position explains substantial variance.

Figure 2 shows the function type distribution across backbones under round-robin topology.

6.3. Backbone × Topology Crossover

Figure 3 visualizes the crossover interaction. Under selector topology, Qwen and Llama partially converge: Qwen's F drops from 232 to 55.8 (R^2: $0.45 \to 0.065$) as its distribution collapses to two types (ACKNOWLEDGMENT $n = 324$, VERIFICATION $n = 567$), while Llama's F rises from 3.81 to 17.3 (R^2: $0.02 \to 0.056$) as it expands to three types (COORDINATION

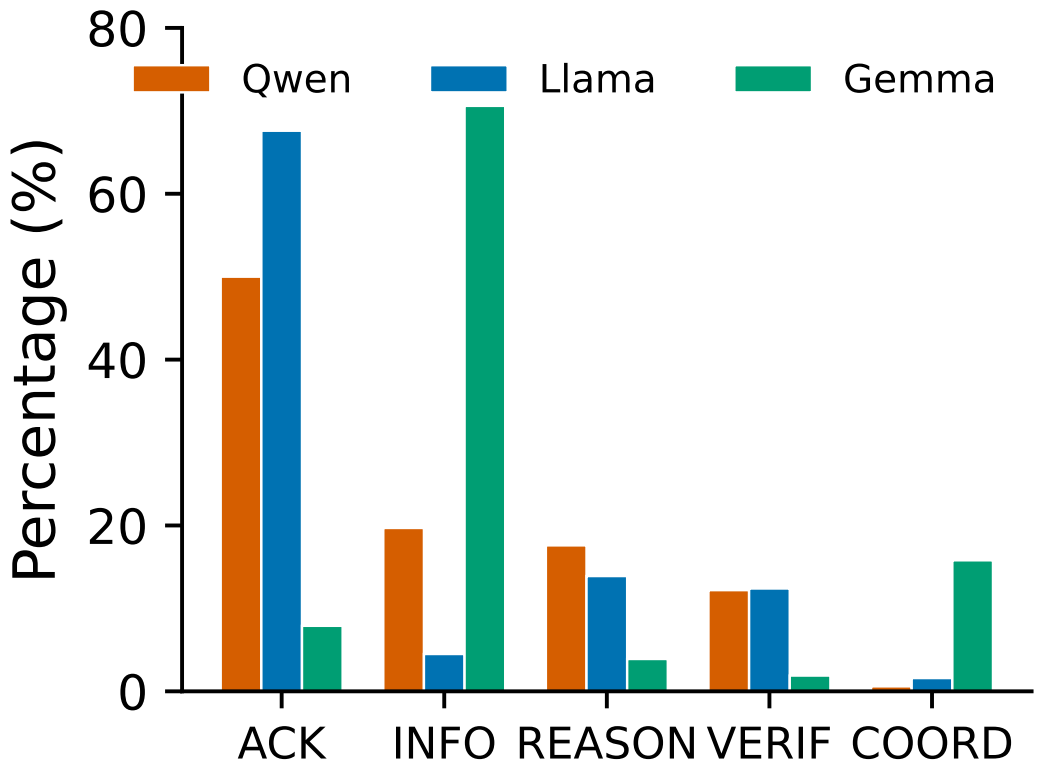


Figure 2: Function type distribution by backbone (round-robin topology). Qwen and Llama are acknowledgment-heavy (>50%), while Gemma is dominated by information provision (70%).

$n = 220$, VERIFICATION $n = 217$, ACKNOWLEDGMENT $n = 428$). Gemma's R^2 drops ($0.34 \rightarrow 0.036$) due to disruption of positional regularity rather than functional collapse, since function type remains non-significant ($F = 0.60$).

6.4. Statistical Validation

Pooled three-way ANOVA. A three-way ANOVA on all $N = 8{,}065$ messages (Table 4) confirms backbone as the dominant factor ($\eta^2 = 0.135$, $F = 629$) and the crossover interaction (backbone $\times$ topology $\eta^2 = 0.043$, $F = 181$). The function $\times$ backbone interaction ($\eta^2 = 0.063$) is 12$\times$ larger than the function main effect ($\eta^2 = 0.005$), reflecting that function type is a strong predictor for Qwen but negligible for Gemma and Llama—pooling dilutes the main effect while preserving the interaction.

Cross-model annotation agreement. Cross-model κ ranges from 0.35 (Gemma) to 0.67 (Qwen), spanning fair-to-substantial agreement [Landis and Koch(1977)]. Qwen's highest κ is consistent with function-driven communication being easier to classify. Gemma falls below our target threshold ($\kappa = 0.35$), indicating weaker cross-model consistency for this backbone. This aligns with its position-driven mode: if function type is not the organizing principle, annotators naturally disagree more on function labels. Per-type κ reveals a complementary pattern: Qwen shows high VERIFICATION agreement ($\kappa \approx 0.66$) but low INFORMATION_PROVISION ($\kappa \approx 0.08$); Gemma shows the reverse ($\kappa \approx 0.87$ and 0.04, respectively). Full agreement tables are in Appendix F.

Task accuracy and surface vs. effective differentiation. Qwen's accuracy is topology-invariant (∼90%) despite dramatic communication changes under selector. Llama presents a paradox: functional differentiation *increases* under selector yet accuracy *drops* (74.8% $\rightarrow$

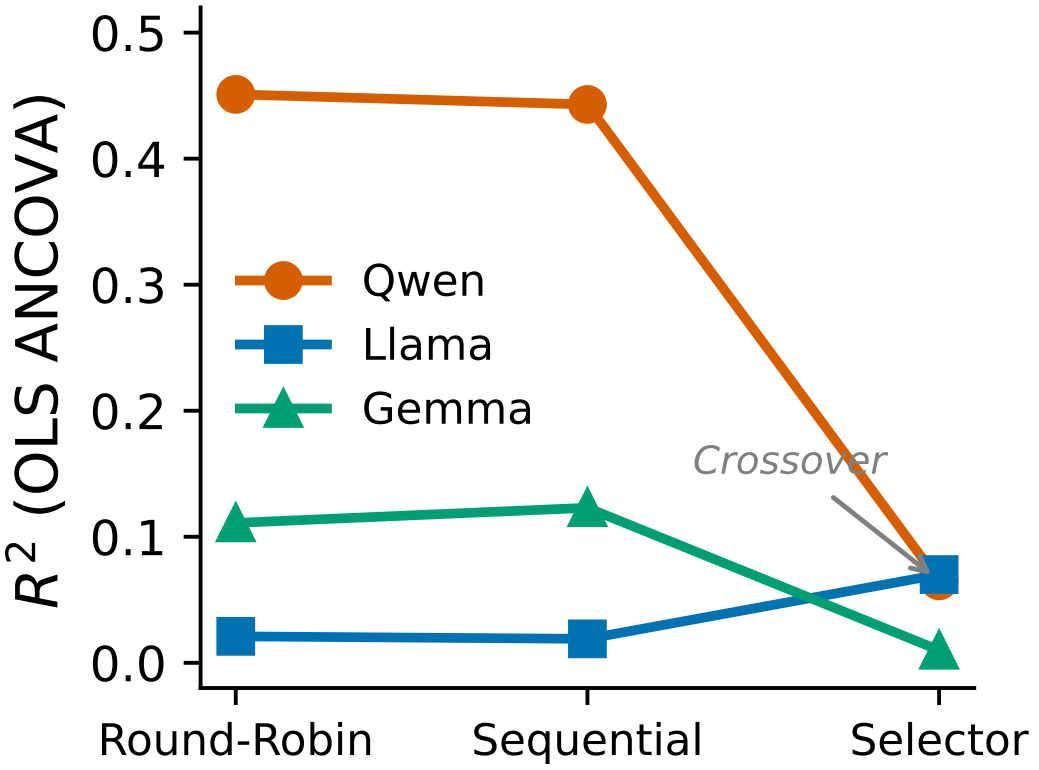


Figure 3: Backbone × topology crossover interaction. R^2 from OLS ANCOVA. Qwen's functional structure degrades under selector while Llama's activates, partially converging at $R^2 \approx 0.06$.

56.2%). This dissociates *surface differentiation* (externally induced diversity) from *effective differentiation* (endogenous specialization that aids task performance). Among Llama's COORDINATION messages under selector, only 7.9% are correct (vs. 83.3% for VERIFICATION); tasks with coordination messages achieve 21.6% success vs. 54.5% without.

7. Discussion

We discuss the implications of our findings and their limitations.

Conversational coherence as structural scaffold. The counter-intuitive impact gradient—acknowledgments most impactful, reasoning guidance least—admits a *conversational coherence* explanation. Acknowledgments are informationally sparse but contextually load-bearing: their absence violates conversational expectations, forcing large behavioral shifts. Reasoning guidance, by contrast, provides content the recipient can reconstruct from the task structure. The neutralize ablation corroborates this: neutralizing (preserving structure, removing content) has the opposite gradient—information-rich types suffer most—confirming that acknowledgment's high remove-impact stems from structural presence rather than informational content (Appendix E). Individual-message ablation reveals that acknowledgments are the least prunable message type: removing a single acknowledgment alters the final answer in 66% of cases (vs. 29% for reasoning guidance). This provides empirical evidence against content-based pruning heuristics [Xing et al.(2025), Zhang et al.(2025c)] and supports the structural scaffold interpretation.

Three communication modes, not a gradient. The ANCOVA decomposition (Table 3) reveals that the impact gradient is not a universal property of multi-agent communication

Table 4: Pooled ANOVA on remove-ablation similarity (N = 8,065, R^2 = 0.298). All $p < .001$.

Term	η^2	F
Backbone	0.135	628.7
Function × Backbone	0.063	67.1
Topology class	0.051	435.4
Backbone × Topology	0.043	181.1
Three-way	0.021	21.1
Function × Topology	0.014	27.7
Function type	0.005	10.4

Table 5: GSM8K accuracy (%) by backbone and topology.

Backbone	**RR**	**SEQ**	**Selector**
Qwen2.5-7B	89.7	90.0	89.7
Gemma-2-9B	80.3	80.7	83.0
Llama-3.1-8B	74.8	71.6	56.2

but depends on backbone-specific communication modes. Qwen is *function-driven* ($F = 232$, $p < 0.001$): function type counterfactually determines downstream impact, producing the sharp gradient in §6.1. Gemma is *position-driven* ($R^2 = 0.34$, but function type $F = 0.43$, n.s.): impact varies with conversational position, not communicative content. Llama is *undifferentiated* ($R^2 = 0.02$): messages have interchangeable effects regardless of type or position. We hypothesize this reflects differences in instruction-following fidelity—models that more precisely modulate output in response to context produce messages whose functional roles are counterfactually distinct, not merely nominally distinct. Whether these modes represent points on a capability continuum or orthogonal strategies requires evaluation across broader model families and scales.

Practical implications. Three design recommendations follow. First, content-based pruning is unreliable: systems should evaluate message prunability through counterfactual probing rather than information-content heuristics. Second, optimal topology depends on communication mode: dynamic routing benefits undifferentiated communicators (activating functional roles) but disrupts function-driven communicators (displacing endogenous structure). The convergence of Qwen and Llama R^2 values under selector (0.065 vs. 0.056) makes this concrete. Third, the surface-vs.-effective differentiation distinction (Llama's selector paradox: more diverse communication, lower accuracy) implies that measuring communication diversity alone is insufficient for evaluating communication quality; evaluation must couple functional analysis with task performance.

Limitations. Our primary experiments use GSM8K; preliminary cross-domain evidence from a 50-task HumanEval pilot ($\kappa = 0.47$; Appendix B) and a MATH validation ($\kappa = 0.45$, function-driven $F = 3.77$, $p = 0.011$; Appendix H) suggest taxonomy applicability across domains, though with reduced effect sizes on harder tasks. Three backbones at 7–9B provide architectural diversity but no scale variation. LLM-based annotation creates potential circularity mitigated by cross-model validation ($\kappa = 0.35$–0.67), though self-annotation bias cannot be fully excluded. Gemma's below-threshold $\kappa = 0.35$ is consistent with its null result on function type ($F = 0.43$, n.s.): if function type does not predict impact, annotators have less structured signal to agree on. Crucially, our core findings (function-driven and undifferentiated modes) rest on Qwen ($\kappa = 0.67$) and Llama ($\kappa = 0.46$–0.63), both above threshold. All primary experiments use greedy decoding; temperature = 0.7 robustness checks confirm stable results for both Qwen ($F = 37.9$, $R^2 = 0.50$) and Llama ($R^2 = 0.10$; Appendix C). All experiments use a single framework (AG2). Our hypotheses (instruction-

following fidelity, structural scaffold) are consistent with data but lack mechanistic evidence from model internals.

8. Conclusion

Counterfactual ablation across 2,693 multi-agent traces reveals that informationally sparse messages—simple acknowledgments—are the most behaviorally load-bearing communications in LLM multi-agent systems, with a single removed acknowledgment altering the final answer in 66% of cases. This counter-intuitive impact gradient is explained by three qualitatively distinct, backbone-dependent communication modes: function-driven (Qwen, $F = 232$), position-driven (Gemma, $R^2 = 0.34$ from position alone), and undifferentiated (Llama, $R^2 = 0.02$). A crossover interaction shows that dynamic routing disrupts endogenous structure in capable models while activating it in weaker ones, implying that optimal topology selection depends on communication mode. These findings challenge content-based pruning heuristics and establish that communication structure in multi-agent systems is an emergent, model-dependent property—a necessary characterization for principled communication optimization.

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A. Qualitative Examples of Communication Functions

Table 6 presents representative messages for each of the five communication function types identified in our taxonomy, selected from annotations with full inter-annotator agreement and maximum confidence scores. Examples are drawn from different backbone–topology combinations to illustrate both the consistency of function definitions and the stylistic variations across models.

[1]This Llama message was classified as REASONING GUIDANCE by both annotators with full agreement, despite its surface resemblance to ACKNOWLEDGMENT. This misclassification illustrates Llama's undifferentiated

Table 6: Qualitative examples for each communication function type. All examples have inter-annotator agreement with confidence = 1.0. Messages are lightly edited for length.

Function Type	**Configuration**	**Message Excerpt**
Reasoning Guidance (2,327 agreed)	Qwen / RR	*"To solve this problem, we need to convert the length of the wire from feet to inches and then divide by the length of each piece. 1. Convert 4 feet to inches: Since 1 foot = 12 inches, 4 feet = $4 \times 12 = 48$ inches. 2. Divide the total length in inches by the length of each piece..."*
	Gemma / Selector	*"Here's how to solve the problem step-by-step: Start: 2 apples. First throw: $2 + 5 = 7$ apples. Second throw: $7 - 1 + 6 = 12$ apples. The answer is 12."*
	Llama / RR	*"I'm glad the solution was accurate. If you have another problem you'd like me to check, feel free to ask."*[1]
Verification (1,001 agreed)	Qwen / RR	*"Step 1: Calculate the initial amount of money Johnny has. Johnny's allowance: \$20. Extra money added: \$10. Total initial amount: \$20 + \$10 = \$30. Step 2: Determine the amount after the investment triples. Tripling the initial amount: \$30 × 3 = \$90. The answer is 90."*
	Llama / RR	*"The given problem is solved step by step. The final answer is 36."*
	Gemma / Selector	*"Here's how to solve the problem: Find the total cost: \$12 (adult) + \$10 (child) = \$22. Add the change: \$22 + \$8 = \$30. Answer: Brittany's mom gave the cashier \$30."*
Acknowledgment (1,854 agreed)	Qwen / RR	*"That is correct. Each single bagel will cost \$1.00 after the discount. The answer is 1."*
	Llama / RR	*"You've correctly summarized the steps and confirmed the calculations. The sequence of events and the final calculation are accurate. The answer is 43."*
	Gemma / RR	*"Your solution is perfectly correct! You clearly laid out the steps and used algebra effectively to find the answer. The answer is 10."*
Information Provision (471 agreed)	Gemma / RR	*"Here's the solution: Sunday's movies: They watch 4 movies / 2 = 2 movies on Sunday. Total weekend movies: 4 + 2 = 6 movies each weekend. Movies in 4 weeks: 6 × 4 = 24 movies. The answer is 24."*
	Qwen / RR	*"To determine how much crepe paper James needs, we first calculate the perimeter of his rectangular backyard fence..."*
	Llama / RR	*"Here's another problem: A bakery is making a special batch of cookies. They need to package 480 cookies into boxes that hold 12 cookies each. How many boxes can they fill with the 480 cookies? Let's solve this problem step by step."*
Coordination (595 agreed)	Llama / RR	*"I'm glad I was able to accurately solve the problem. If you have any more math problems you'd like me to solve, feel free to ask."*
	Gemma / RR	*"Please provide me with the problem you'd like me to solve! I'm ready to check the solver's solution..."*
	Llama / Selector	*"I'm glad you were able to provide the correct steps and solution. You have a good understanding of the problem..."*

B. HumanEval Cross-Domain Pilot

To provide preliminary evidence of cross-domain applicability, we ran a 50-task pilot using Qwen2.5-7B with a hierarchical (solver–reviewer) topology on the HumanEval code generation benchmark [Chen et al.(2021)Chen, Tworek, Jun, Yuan, Pinto, Kaplan, Edwards, Burda, Joseph, Brockman, et al.].

Results. The system achieved 80% pass@1 accuracy (40/50 tasks). The communication function distribution differs markedly from GSM8K: REASONING_GUIDANCE dominates (50.8%), followed by VERIFICATION (41.5%), with ACKNOWLEDGMENT (4.2%), COORDINATION (1.7%), and INFORMATION_PROVISION (1.7%) appearing rarely. Cross-model annotation agreement reached $\kappa = 0.47$ (moderate), comparable to the lower end of the GSM8K range.

Limitations. The hierarchical topology proved too efficient for statistical analysis: 86% of tasks were solved in a single round (2 messages), yielding only 118 total messages and 18 eligible for local impact analysis. Most function types had $n < 30$, precluding Kruskal–Wallis and OLS ANCOVA tests. The distributional shift (from acknowledgment-heavy in GSM8K to reasoning-guidance-heavy in HumanEval) is itself informative—it suggests that the taxonomy captures domain-specific communication patterns—but the statistical framework requires substantially more multi-round conversations to produce meaningful results in this domain.

C. Temperature Robustness

To assess sensitivity to decoding temperature, we ran a 50-task subset of Qwen × round-robin at temperature = 0.7 (primary experiments use greedy decoding). Accuracy remains stable (90.0% vs. 89.7%). The Kruskal–Wallis test remains significant ($H = 34.0$, $p < 0.001$), and OLS ANCOVA confirms function-driven communication ($F = 37.9$, $p < 0.001$; $R^2 = 0.50$; cross-model $\kappa = 0.60$). A parallel check on Llama × round-robin at temperature = 0.7 yields $R^2 = 0.10$, $\kappa = 0.41$, consistent with its undifferentiated mode under greedy decoding. The function distribution shifts toward more REASONING_GUIDANCE (37% vs. 18%) and less ACKNOWLEDGMENT (36% vs. 50%), suggesting stochastic decoding produces more substantive content without altering the functional structure.

D. OLS Robustness Checks

We validate the OLS ANCOVA (Equation 2) with three robustness checks across all nine configurations.

Residual diagnostics. Shapiro–Wilk tests reject normality for all configurations (expected at $n > 400$); Breusch–Pagan tests detect heteroscedasticity in all cases. We therefore report HC3 robust standard errors throughout the main text. All conclusions are unchanged under robust inference. VIF analysis shows high collinearity between `message_position` and `round_number` (VIF > 10), but function type dummy VIF values remain below 3, confirming that multicollinearity does not affect the key func_type inference. Q-Q plots are available in the supplementary materials.

Nonlinear position model. We augment the linear model with a quadratic position term (`position`^{2}). Table 7 reports key comparisons. For Qwen and Llama, adding `position`^{2} produces negligible $\Delta R^2 < 0.005$ and function type remains significant ($p < 0.001$). For Gemma, `position`^{2} absorbs $\Delta R^2 = 0.228$ (RR) and 0.215 (SEQ), rendering function type non-significant ($p > 0.69$). This confirms that Gemma's position-driven mode reflects genuine nonlinear positional dynamics rather than a linear-model artifact.

Message length control. Adding message length (character count) as a covariate produces $\Delta R^2 < 0.04$ for all configurations. Function type remains significant for Qwen ($p < 0.001$) and Llama ($p < 0.03$) after controlling for length, confirming that the impact gradient is not an artifact of message length differences across function types.

communication mode (Section 7): its messages are functionally interchangeable regardless of nominal category, making even high-confidence annotations unreliable for this backbone.

Table 7: Linear vs. quadratic position model: function type significance after controlling for nonlinear position effects.

Configuration	Linear R^2	Quad R^2	ΔR^2	func_type p (quad)
Qwen RR	.451	.452	.001	$< .001$
Qwen SEQ	.443	.447	.004	$< .001$
Qwen SEL	.065	.065	.000	$< .001$
Gemma RR	.111	.339	**.228**	.806
Gemma SEQ	.123	.338	**.215**	.699
Gemma SEL	.010	.040	.031	.219
Llama RR	.021	.022	.001	.010
Llama SEQ	.019	.020	.001	.004
Llama SEL	.070	.074	.004	$< .001$

E. Neutralize Ablation Analysis

While the main text reports remove-ablation impact, neutralize ablation (replacing a message with "`[message removed]`" while preserving its slot) measures a complementary dimension. The two ablation types produce *opposite* impact gradients:

- **Remove** (content + structure removed): acknowledgment most impactful (sim = 0.35), reasoning guidance least (0.78).
- **Neutralize** (content removed, structure preserved): information provision most impactful (sim = 0.05), verification least (0.50).

The difference $\Delta = \text{sim}_{\text{remove}} - \text{sim}_{\text{neutralize}}$ isolates the *structural value* of each message type. Information provision shows the largest $\Delta = 0.47$ (almost entirely content value), while coordination shows $\Delta = 0.07$ (almost entirely structural value). This decomposition confirms that acknowledgment's high remove-impact is driven by structural presence—violating conversational expectations—rather than informational content.

F. Supplementary Tables

Table 8 reports the full cross-model annotation agreement matrix.

Table 8: Cross-model annotation agreement (Cohen's κ) by backbone and topology.

Backbone	Round-Robin	Sequential	Selector
Qwen2.5-7B	0.64	0.64	0.669
Gemma-2-9B	0.350	0.348	0.351
Llama-3.1-8B	0.46	0.47	0.634

G. Impact Metric Robustness

To verify that the impact gradient (Section 6.3) is not an artifact of the MiniLM embedding metric, we recompute message impact using three independent text similarity measures: ROUGE-L, token-level F1, and Jaccard similarity. Table 9 reports the Spearman rank correlation (ρ) between MiniLM cosine similarity and each alternative metric across all nine configurations.

For the primary Qwen round-robin configuration, all four metrics produce identical function type impact rankings ($\rho = 1.0$): COORDINATION < ACKNOWLEDGMENT < INFORMATION_PROVISION < VERIFICATION < REASONING_GUIDANCE. Across all nine configurations, 7 of 9 show perfectly consistent gradient ordering; the remaining two (Llama RR, Gemma SEQ) differ only in adjacent categories. The mean $\rho > 0.91$ across all metric pairs confirms that the impact gradient is a property of the underlying behavioral change, not an artifact of the embedding model.

Table 9: Spearman ρ between MiniLM cosine similarity and alternative impact metrics. All correlations computed over per-message impact scores within each configuration.

Configuration	ROUGE-L	Token F1	Jaccard
Qwen RR	1.0	1.0	1.0
Qwen SEQ	1.0	1.0	1.0
Qwen SEL	1.0	1.0	1.0
Gemma RR	1.0	1.0	1.0
Gemma SEQ	1.0	0.8	0.8
Gemma SEL	0.949	0.949	0.949
Llama RR	0.6	0.6	0.6
Llama SEQ	1.0	1.0	1.0
Llama SEL	0.9	0.9	0.9
Mean	**0.939**	**0.917**	**0.917**

H. Cross-Domain Validation: MATH

To test whether the three communication modes generalize beyond elementary arithmetic, we ran the full pipeline on 50 MATH tasks (Level 3–5, Algebra and Number Theory) using the same round-robin topology.

Qwen on MATH. Accuracy is 72.0% (36/50), lower than GSM8K (89.7%) as expected for harder tasks. Each task produces 6 messages (300 total, 250 eligible for analysis after excluding initial prompts). Cross-model annotation agreement reaches $\kappa = 0.45$ (moderate, above threshold). The OLS ANCOVA yields $R^2 = 0.098$ with function type significant after position control ($F = 3.77$, $p = 0.011$), confirming that Qwen's function-driven mode persists on a harder mathematical domain, though with reduced effect size ($R^2 = 0.098$ vs. 0.45 on GSM8K). The Kruskal–Wallis test is borderline ($H = 7.37$, $p = 0.061$), likely due to reduced sample size.

Gemma on MATH. Accuracy drops to 40.0% (20/50). Cross-model $\kappa = 0.48$ (above threshold). The OLS ANCOVA yields $R^2 = 0.236$ but function type is non-significant ($F = 0.09$, $p = 0.76$), while message position is highly significant ($F = 15.22$, $p < 0.001$). This replicates Gemma's position-driven mode from GSM8K: on MATH, position again explains the variance while function type does not. The Kruskal–Wallis test is also non-significant ($H = 2.05$, $p = 0.15$).

Llama on MATH. Accuracy is 64.3% (27/42; 8 tasks produced empty traces). Cross-model $\kappa = 0.36$ (below threshold). With only 168 total messages, ACKNOWLEDGMENT dominates (56%), and most other function types fall below $n = 30$, precluding OLS ANCOVA and Kruskal–Wallis analysis. The acknowledgment-dominated distribution is directionally consistent with Llama's undifferentiated mode on GSM8K, though statistical confirmation requires larger samples.

Summary. Qwen's function-driven mode ($F = 3.77$, $p = 0.011$) and Gemma's position-driven mode ($F = 0.09$, n.s.; position $F = 15.22$, $p < 0.001$) both replicate on MATH. Llama's distribution is directionally consistent with its undifferentiated mode but lacks statistical power. Two of three communication modes are confirmed to generalize beyond elementary arithmetic.

I. Annotation Prompt

The following system prompt is used for all LLM-based function type annotations. Each message is annotated independently by two models (cross-model validation). Temperature is set to 0 with constrained JSON output.

```
You are a communication analyst for multi-agent systems. Your task is to classify
the function of a message in a multi-agent math problem-solving conversation.

Function Types

1. information_provision: Provides facts, data, intermediate calculations,
or partial results that advance the solution. Includes reporting computed
values, stating known quantities, or presenting new numerical results.

2. reasoning_guidance: Provides reasoning direction, problem decomposition,
step-by-step plans, or conceptual frameworks. Includes breaking the problem
into sub-problems, suggesting solution strategies, or outlining multi-step
approaches.

3. verification: Checks, corrects, or confirms the correctness of previous
work. Includes identifying errors, validating results, pointing out mistakes,
or confirming a step is correct with a specific reason.

4. coordination: Manages the flow of the conversation, assigns tasks, requests
clarification, or controls turn-taking. Includes delegating sub-tasks, asking
for more information, or directing the problem-solving process.

5. acknowledgment: Simple confirmation, repetition, or agreement that adds
no new information or reasoning. Includes restating what was already said
without extending it.

Decision Rules

- If a message both provides new results AND checks previous work, classify
as verification.
- If a message provides a reasoning plan AND new results, classify as reasoning_guidance.
- If a message confirms correctness WITH a substantive explanation, classify
as verification.
- If a message simply agrees or restates without adding new information, classify
as acknowledgment.

Output Format

Respond with ONLY a JSON object:
{''type'': ''<function_type>'', ''confidence'': <0.0-1.0>, ''reasoning'': ''<brief
explanation>''}
```

The user prompt provides the conversation context with all messages and marks the target message for classification:

```
## Conversation Context
[Message 0] {sender} -> {receiver}:
{content}
[Message 1] {sender} -> {receiver}: << TARGET MESSAGE >>
{content}
[Message 2] {sender} -> {receiver}:
{content}
...
Classify the function of Message {target_idx}.
```

NeurIPS Paper Checklist

1. **Claims Yes**: All claims are supported by empirical evidence from our experiments. We have downgraded language from "causal" to "counterfactual" to accurately reflect our ablation methodology.

2. **Limitations Yes**: Section 7 discusses limitations including single task domain, single scale, LLM-based annotation circularity, self-annotation bias, and greedy decoding.

3. **Theory Assumptions and Proofs N/A**: Our work is empirical; no theoretical proofs are presented.

4. **Experimental Result Reproducibility Yes**: Section 4–5 specify all models, hyperparameters, framework versions, and statistical tests. Greedy decoding ensures deterministic outputs. Code and data will be released upon acceptance.

5. **Open access to data and code Yes**: Code, traces, and annotations will be released upon acceptance under an open-source license. GSM8K is publicly available.

6. **Experimental Setting/Details Yes**: Sections 4–5 provide full experimental details including model names, parameter counts, generation settings, annotation protocol, and statistical framework.

7. **Compute Resources Yes**: Experiments were conducted on 4×RTX 6000 Pro (96GB each) GPUs. Total compute is modest: all 9 configurations complete in approximately 24 hours.

8. **Code Of Ethics Yes**: This research studies existing open-weight models on a public benchmark. No new models are trained. No human subjects are involved.

9. **Broader Impacts Yes**: Our work characterizes communication in multi-agent systems to improve efficiency and interpretability. We do not foresee negative societal impacts beyond those inherent to LLM deployment generally.

10. **Safeguards N/A**: Our work analyzes existing model behaviors on mathematical reasoning tasks and does not introduce new risks.

11. **Licenses Yes**: GSM8K is MIT-licensed. All three model backbones use permissive open-weight licenses. AG2/AutoGen is Apache 2.0.

12. **New Assets Yes**: We release interaction traces, function type annotations, and ablation results. These will be documented and released under an open license upon acceptance.

13. **Crowdsourcing and Research with Human Subjects N/A**: No human subjects or crowdsourcing were involved.

14. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects N/A**: No human subjects research.